

Module -4

Fourier Transform

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Fourier Transform

- The Fourier transform is an integral transform like the Laplace transform.
- It transforms a function in the time domain into the frequency domain.
- The Fourier transform is very useful in communication systems and digital signal processing, in situations where the Laplace transform does not apply.
- The Laplace transform handle circuits with inputs for $t > 0$ with initial conditions, whereas Fourier transform can handle with inputs for $t > 0$ and $t < 0$.
- The Fourier transform allows us to extend the concept of a frequency spectrum to non-periodic functions.

Definition of Fourier Transform

The **Fourier transform** is an integral transformation of $f(t)$ from the time domain to the frequency domain.

$$F(\omega) = \mathcal{F}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-j\omega t} dt$$

- $F(\omega)$ is a complex function;
- Its magnitude is called the amplitude spectrum
- Its phase is called the phase spectrum
- Thus $F(\omega)$ is the spectrum

Inverse Fourier Transform

$$f(t) = \mathcal{F}^{-1}[F(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

- The Fourier transform exists when the Fourier integral converges and the sufficient condition required is given below:

$$\int_{-\infty}^{\infty} |f(t)| dt < \infty$$

Fourier Transform pairs

$f(t)$	$F(\omega)$
$\delta(t)$	1
1	$2\pi\delta(\omega)$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$u(t + \tau) - u(t - \tau)$	$2\frac{\sin\omega\tau}{\omega}$
$ t $	$\frac{-2}{\omega^2}$
$\text{sgn}(t)$	$\frac{2}{j\omega}$
$e^{-at}u(t)$	$\frac{1}{a + j\omega}$
$\delta(t - t_0)$	$e^{-j\omega t_0}$

$f(t)$	$F(\omega)$
$e^{at}u(-t)$	$\frac{1}{a - j\omega}$
$t^n e^{-at}u(t)$	$\frac{n!}{(a + j\omega)^{n+1}}$
$e^{-a t }$	$\frac{2a}{a^2 + \omega^2}$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\sin \omega_0 t$	$j\pi[\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
$\cos \omega_0 t$	$\pi[\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
$e^{-at} \sin \omega_0 t u(t)$	$\frac{\omega_0}{(a + j\omega)^2 + \omega_0^2}$
$e^{-at} \cos \omega_0 t u(t)$	$\frac{a + j\omega}{(a + j\omega)^2 + \omega_0^2}$

Circuit Applications

- The Fourier transform is applied to circuits with non-periodic sinusoidal excitations in the same way as phasor techniques.

$$V(\omega) = Z(\omega)I(\omega)$$

- Where $V(\omega)$ – FT of Voltage, $I(\omega)$ – FT of current and $Z(\omega)$ – Impedance.

R	$\Rightarrow$	R
L	$\Rightarrow$	$j\omega L$
C	$\Rightarrow$	$\frac{1}{j\omega C}$

Steps for solving circuits using FT

- Transform the circuit elements into frequency domain and take the Fourier transform of the excitations.
- Use any circuit solving technique to find the response in frequency domain.
- Take the inverse Fourier transform to get the response in time domain.

Transfer Function

- Although the Fourier transform method produces a response that exists for $-\infty < t < \infty$, Fourier analysis cannot handle circuits with initial conditions.
- The transfer function is again defined as the ratio of the output response $Y(\omega)$ to the input excitation $X(\omega)$.

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$



$$Y(\omega) = H(\omega)X(\omega)$$

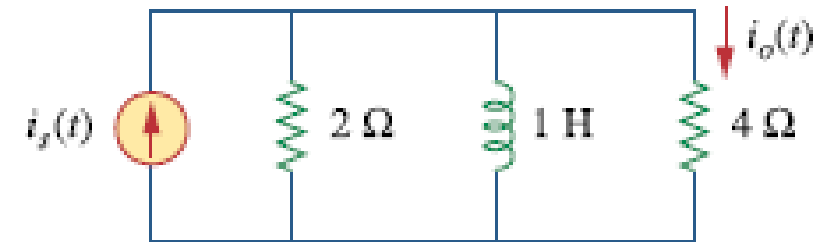
- $H(\omega)$ is identical to $H(s)$ with $s = j\omega$.
- If $x(t) = \delta(t)$, then $X(\omega) = 1$, so that

$$Y(\omega) = H(\omega) = \mathcal{F}[h(t)]$$

- $h(t)$ is the impulse response.

Numerical - 1

Find the transfer function $I_o(\omega)/I_s(\omega)$ for the circuit



Solution

$$2 \parallel j\omega = \frac{j2\omega}{2 + j\omega}$$

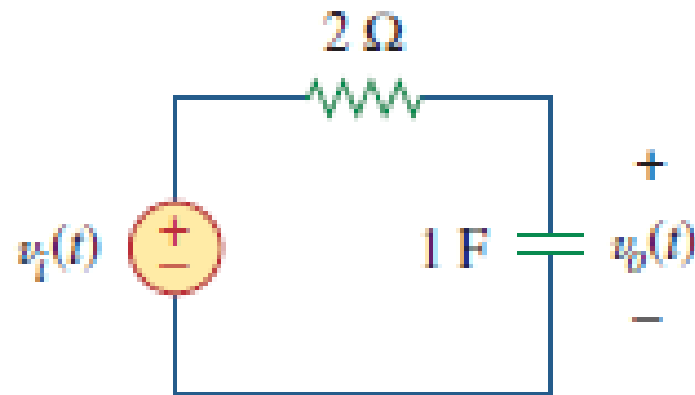
By current division,

$$H(\omega) = \frac{I_o(\omega)}{I_s(\omega)} = \frac{\frac{j2\omega}{2 + j\omega}}{4 + \frac{j2\omega}{2 + j\omega}} = \frac{j2\omega}{j2\omega + 8 + j4\omega}$$

$$H(\omega) = \underline{\underline{\frac{j\omega}{4 + j3\omega}}}$$

Numerical - 2

Find $v_o(t)$ in the circuit of Fig. for $v_i(t) = 2e^{-3t}u(t)$.



Solution:

The Fourier transform of the input voltage is

$$V_i(\omega) = \frac{2}{3 + j\omega}$$

and the transfer function obtained by voltage division is

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = \frac{1/j\omega}{2 + 1/j\omega} = \frac{1}{1 + j2\omega}$$

Hence,

$$V_o(\omega) = V_i(\omega)H(\omega) = \frac{2}{(3 + j\omega)(1 + j2\omega)}$$

or

$$V_o(\omega) = \frac{1}{(3 + j\omega)(0.5 + j\omega)}$$

By partial fractions,

$$V_o(\omega) = \frac{-0.4}{3 + j\omega} + \frac{0.4}{0.5 + j\omega}$$

Taking the inverse Fourier transform yields

$$v_o(t) = 0.4(e^{-0.5t} - e^{-3t})u(t)$$

Numerical - 3

The transfer function of a circuit is

$$H(\omega) = \frac{2}{j\omega + 2}$$

If the input signal to the circuit is $v_s(t) = e^{-4t}u(t)$ V find the output signal. Assume all initial conditions are zero.

Solution

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} \longrightarrow Y(\omega) = H(\omega)X(\omega)$$

$$x(t) = v_s(t) = e^{-4t}u(t) \longrightarrow X(\omega) = \frac{1}{4 + j\omega}$$

$$Y(\omega) = \frac{2}{(j\omega + 2)(4 + j\omega)} = \frac{2}{(s + 2)(s + 4)}, \quad s = j\omega$$

$$Y(s) = \frac{A}{s + 2} + \frac{B}{s + 4}$$

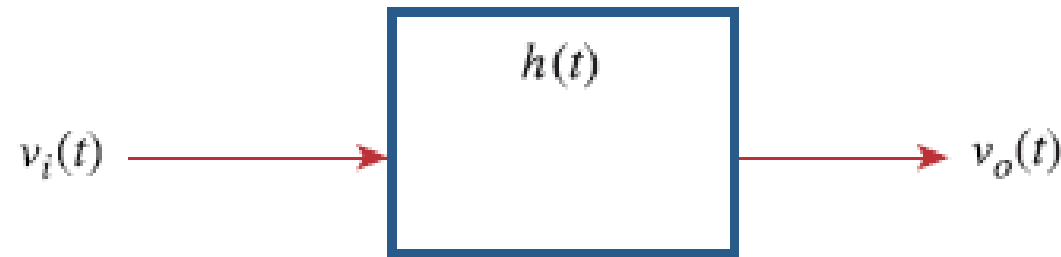
$$A = \frac{2}{-2 + 4} = 1, \quad B = \frac{2}{-4 + 2} = -1$$

$$Y(s) = \frac{1}{s + 2} - \frac{1}{s + 4}$$

$$y(t) = \underline{\left(e^{-2t} - e^{-4t} \right) u(t)}$$

Numerical - 4

For the linear system when the input voltage is $v_i(t) = 2\delta(t)$ V, the output is $v_o(t) = 10e^{-2t} - 6e^{-4t}$ V. Find the output when the input is $v_i(t) = 4e^{-t}u(t)$ V.



Solution

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = \frac{\frac{10}{2 + j\omega}}{2} - \frac{6}{4 + j\omega} = \frac{5}{2 + j\omega} - \frac{3}{4 + j\omega}$$

$$V_o(\omega) = H(\omega)V_i(\omega) = \left(\frac{5}{2+j\omega} - \frac{3}{4+j\omega} \right) \frac{4}{1+j\omega}$$

$$= \frac{20}{(s+1)(s+2)} - \frac{12}{(s+1)(s+4)}, \quad s = j\omega$$

Using partial fraction,

$$V_o(\omega) = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+1} + \frac{D}{s+4} = \frac{16}{1+j\omega} - \frac{20}{2+j\omega} + \frac{4}{4+j\omega}$$

Thus,

$$v_o(t) = \underline{\left(16e^{-t} - 20e^{-2t} + 4e^{-4t} \right) u(t) \text{ V}}$$

Numerical - 5

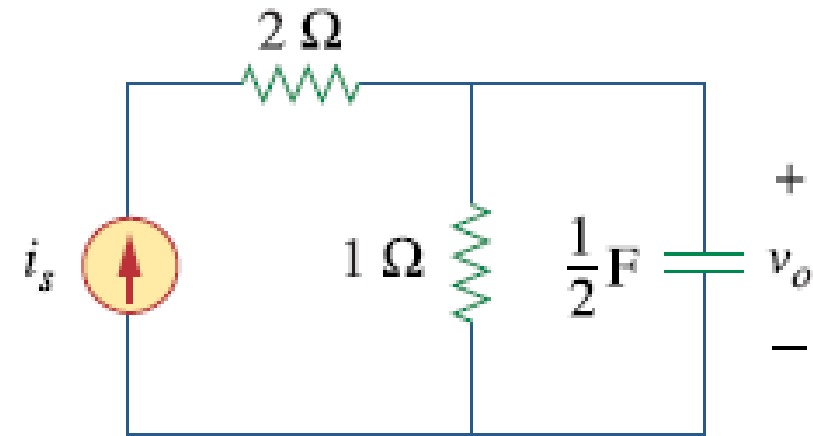
Find the voltage $v_o(t)$ in the circuit

Let $i_s(t) = 8e^{-t}u(t)$ A.

Solution

$$\frac{1}{2}F \longrightarrow \frac{1}{j\omega C} = \frac{2}{j\omega}$$

$$I_o = \frac{1}{1 + \frac{2}{j\omega}} I_s$$



$$V_o = \frac{2}{j\omega} I_o = \frac{\frac{2}{j\omega}}{1 + \frac{2}{j\omega}} I_s = \frac{2}{2 + j\omega} \frac{8}{1 + j\omega}$$

$$= \frac{16}{(s+1)(s+2)}, s = j\omega$$

$$= \frac{A}{s+1} + \frac{B}{s+2}$$

where $A = 16/1 = 16$, $B = 16/(-1) = -16$

Thus,

$$v_o(t) = \underline{\underline{16(e^{-t} - e^{-2t})u(t) \text{ V.}}}$$